

Performance of Packet-Switched Networks (Desempenho de Redes com Comutação de Pacotes)

Modelação e Desempenho de Redes e Serviços

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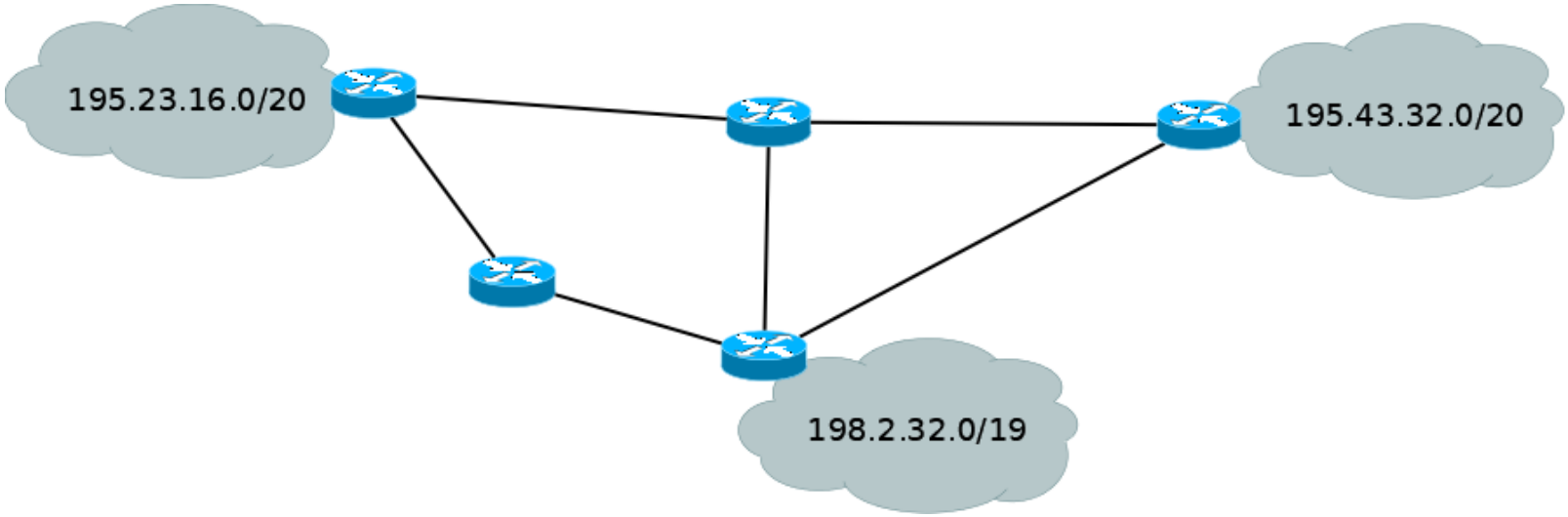
DETI-UA, 2026/2027

Routing in Packet-Switched Networks

Packet-switched networks are categorized into two main types:

- Connection-Oriented (Virtual Circuit) Packet Switching
- Connectionless (Datagram) Packet Switching

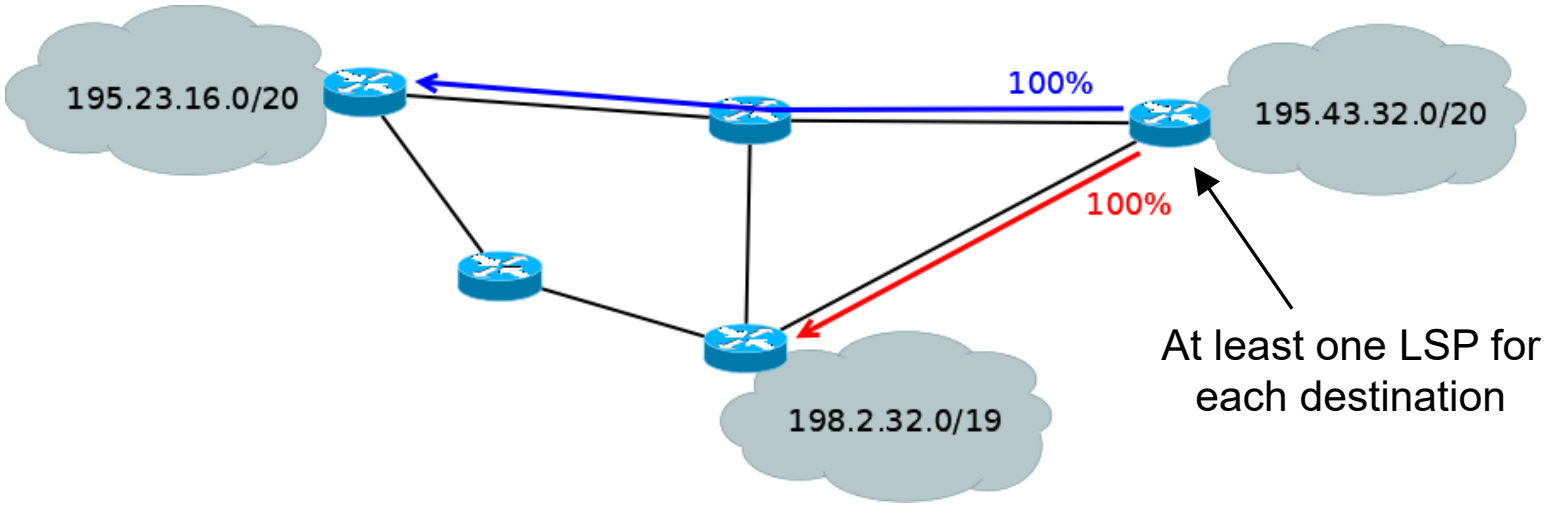
Consider the following very small example of an ISP (*Internet Service Provider*) core network connecting 3 access networks (networks with attached user clients) with assigned IP address aggregates:



Routing in
connection-oriented packet switching networks

- Each packet flow is assigned with at least one logical routing path (named virtual circuit).
- The virtual circuits are initially established.
- Packet transmission is only possible after the establishment of the virtual circuits.

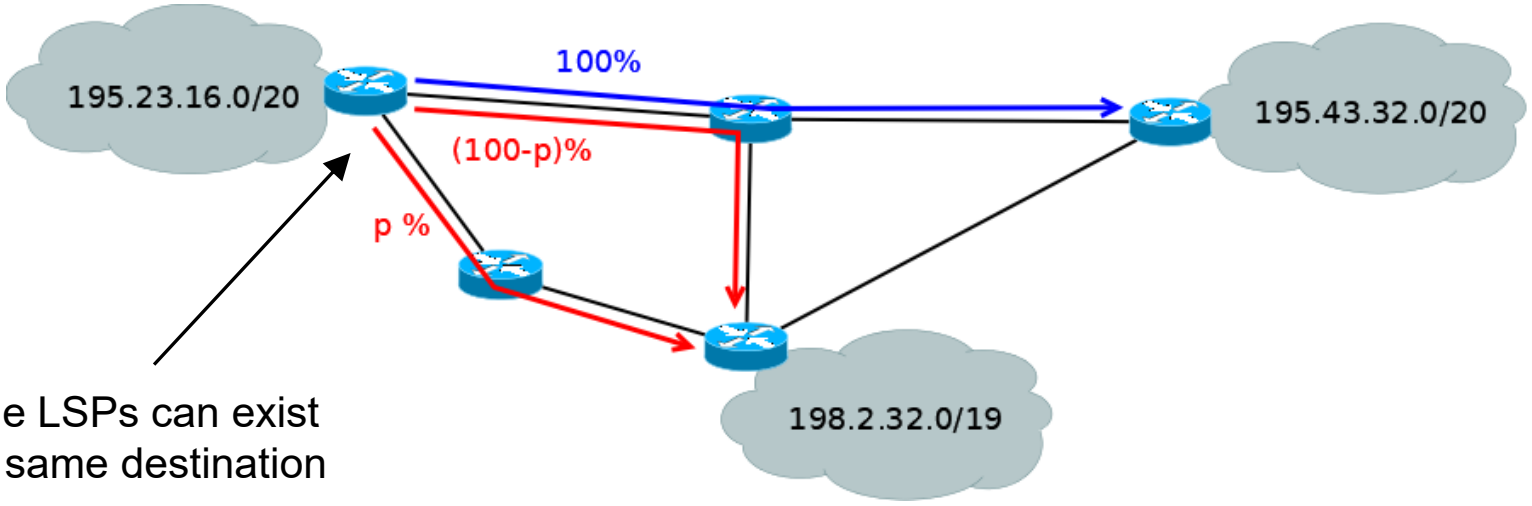
Example: IP/MPLS networks, where virtual circuits are named LSPs (*Label Switched Paths*).



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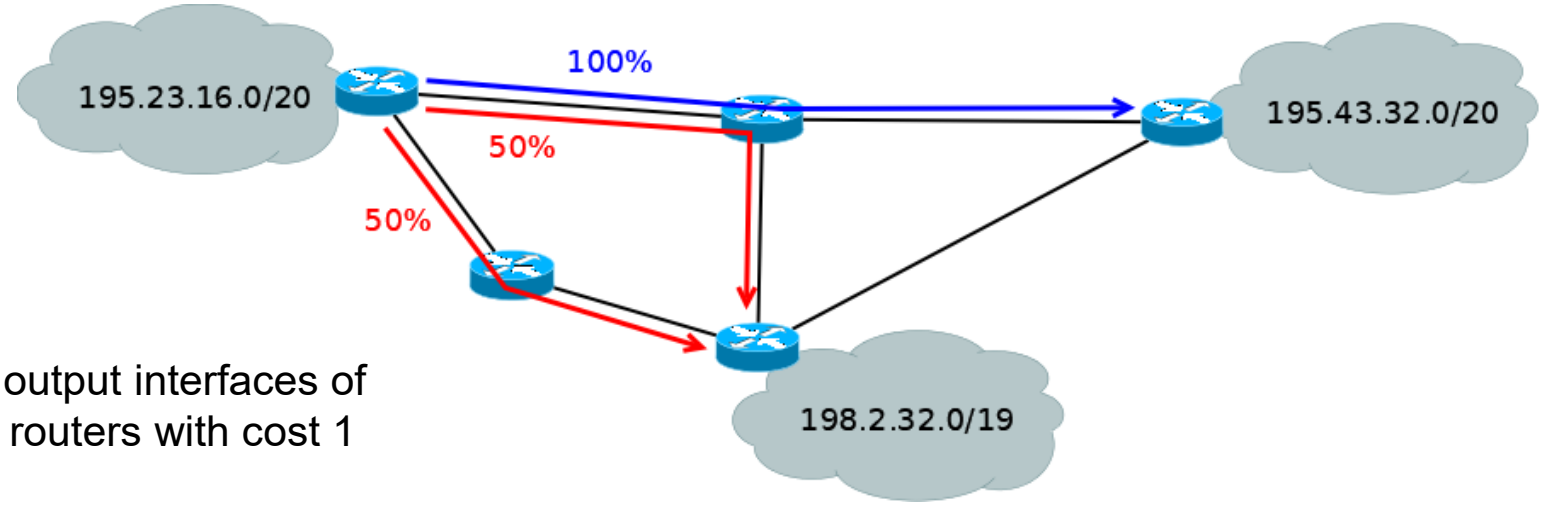
Example: IP/MPLS networks, where virtual circuits are named LSPs (*Label Switched Paths*).



Routing in
connectionless packet switching networks

- Each router output interface is assigned with a cost value.
- The routing path cost (from a router to a destination network) is the sum of the costs of the output interfaces in the routing path.
- On each router, each packet is forwarded by the minimum cost path to the destination.
- When there are multiple minimum cost paths, the ECMP (Equal Cost Multi-Pathing) technique is used.

Examples: IP networks with routing protocols such as OSPF or IS-IS.

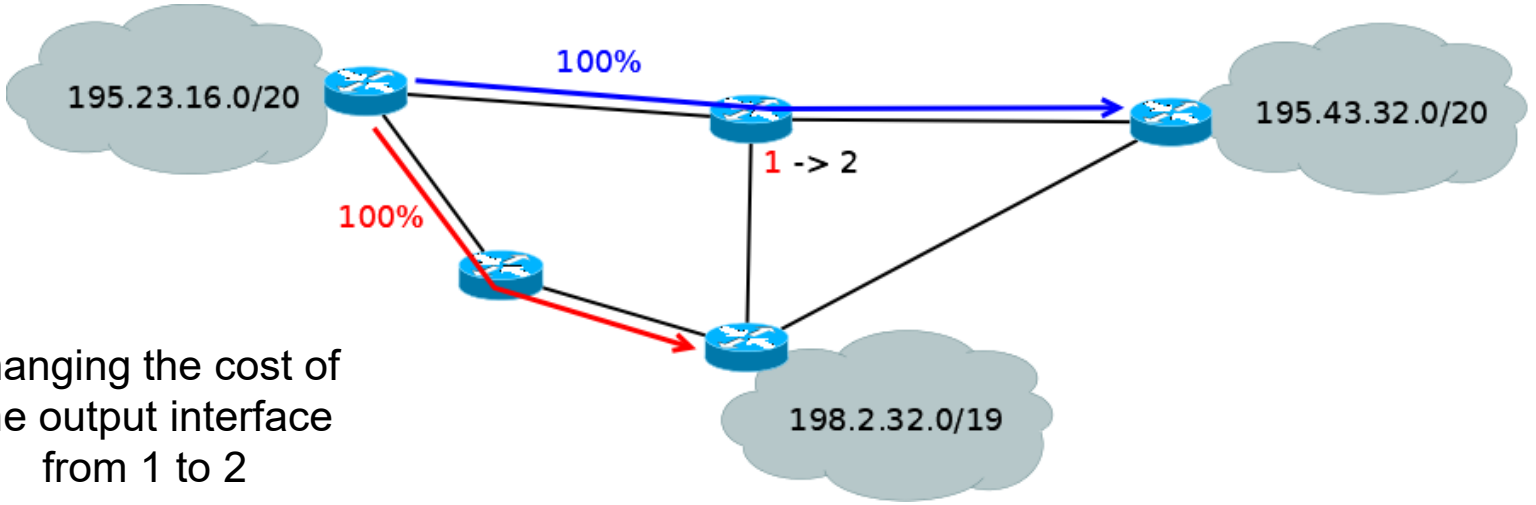


All output interfaces of all routers with cost 1

Routing in
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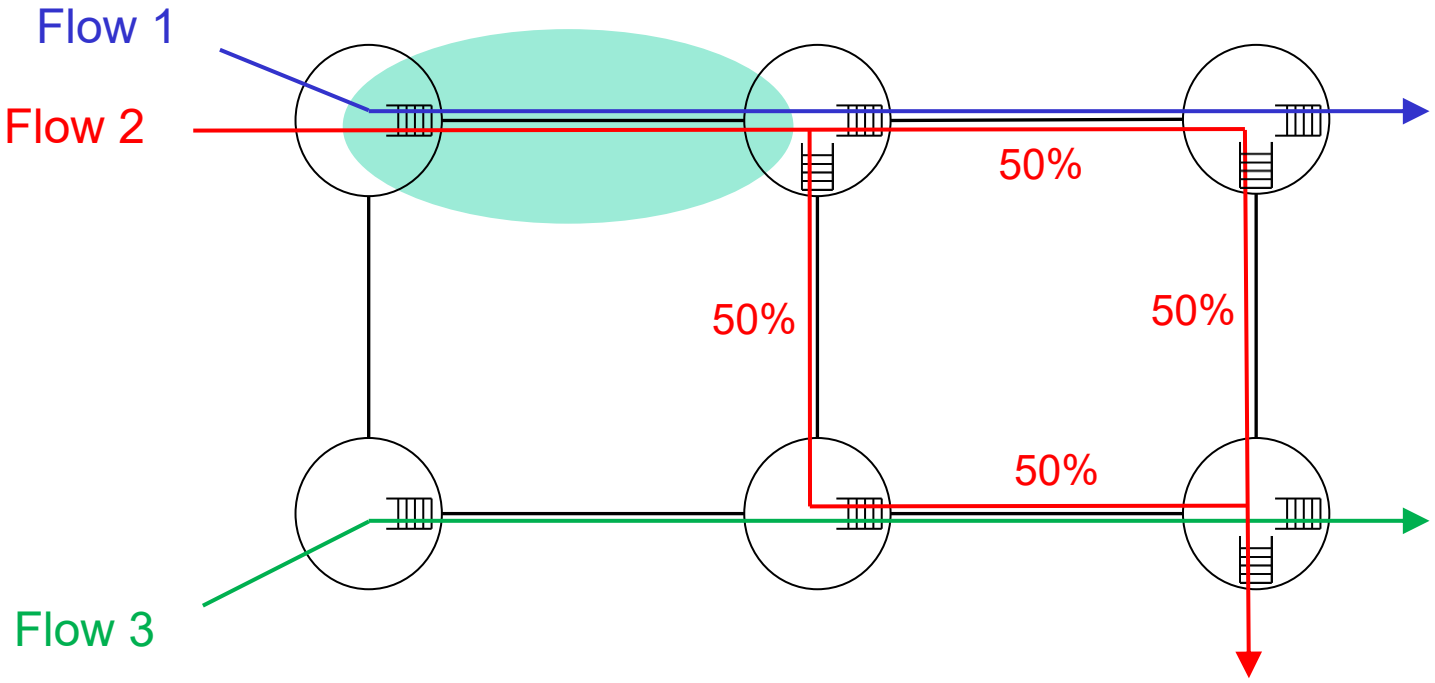
Changing the cost of one output interface from 1 to 2

Modelling the routing in packet-switched network

A network is modelled by a set of nodes (representing routers) and a set of existing links connecting nodes.

Routing defines the sequence of links through which the packets of each flow and each service are forwarded from the source to the destination node.

We have already analysed a single link: *M*/*G*/1, *M*/*M*/1, *M*/*M*/1/*m*, *ber*, priority queueing, round robin schedulers

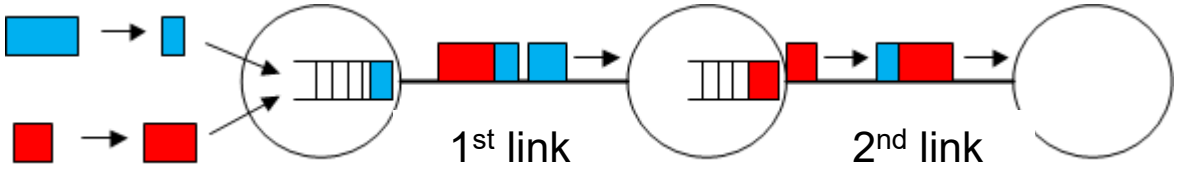


Routing path with multiple links

In a routing path with multiple links, the time intervals between consecutive packet arrivals are correlated with the size of the packets transmitted in the previous link. This fact makes its modelling harder.

Consider the example of the figure:

- the routing path has two links;
- the packet arrivals of the two flows are Poisson processes, and the packet size of both flows is exponentially distributed with the same average size.



In this case, the 1st link is a *M*/*M*/1 queuing system.

However, the 2nd link is not a *M*/*M*/1 queuing system because:

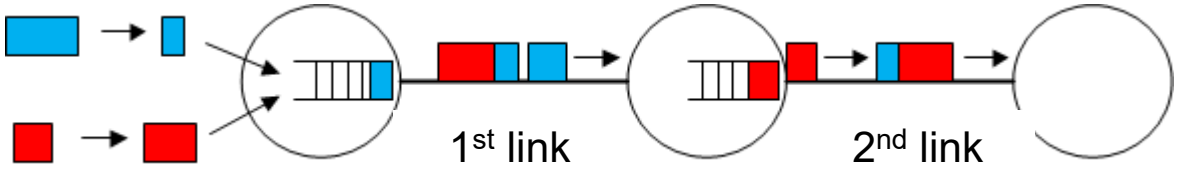
- the interarrival time between consecutive packets in the 2nd link cannot be shorter than the transmission time of the second packet in the 1st link;
- so, the packet interarrival time is not exponentially distributed and, therefore, packet arrivals are not Poisson processes in the 2nd link.

Kleinrock approximation

In a routing path with multiple links, the time intervals between consecutive packet arrivals are correlated with the size of the packets transmitted in the previous link. This fact makes its modelling harder.

Consider the example of the figure:

- the routing path has two links;
- the packet arrivals of the two flows are Poisson processes, and the packet size of both flows is exponentially distributed with the same average size.



The *Kleinrock approximation* consists in assuming that packet arrivals of all flows are Poisson processes in all links of their routing paths (i.e. it ignores the correlation between packet sizes and packet interarrival times)

- Kleinrock approximation only affects the system modelling in congested time periods (when packets are being transmitted without any time interval between them).
- So, the shorter and less frequent the congested time periods are, the better the Kleinrock approximation is.

Average packet delay of unicast services

- The traffic of a unicast service s is defined by a set of flows $f = 1 \ldots F_s$ (F_s is the no. of flows of service s) such that each flow is defined by:
  - a given packet arriving rate,
  - from a given source node of the network,
  - to a given destination node of the network.
- If for each flow $f = 1 \ldots F_s$:
  - λ_f (in pps) is the packet arriving rate at its source node
  - W_f (in seconds) is the average packet delay suffered on its routing path(s) in the network (from its origin to its destination node)

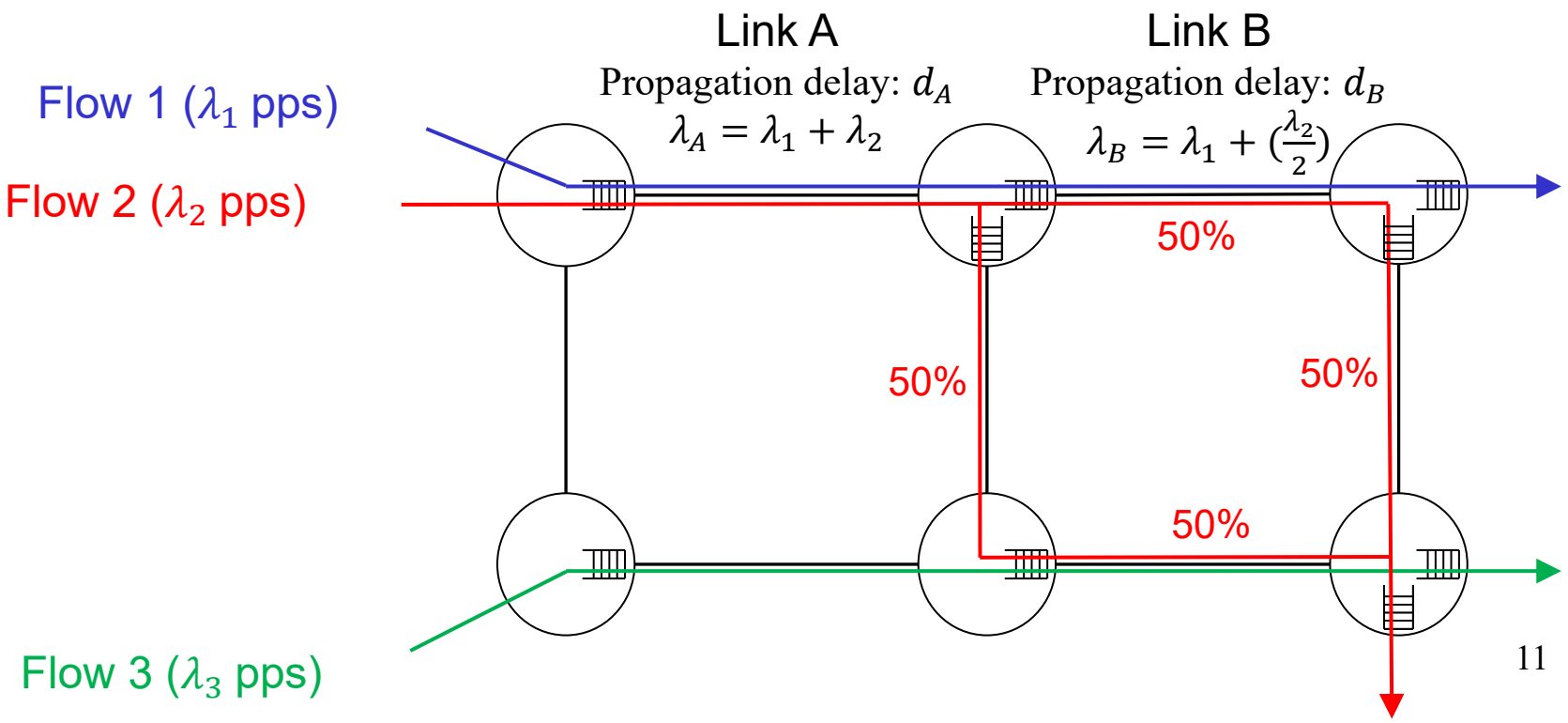
  then, the service average packet delay W_s is the weighted average: $$W_s = \frac{\sum_{f=1}^{F_s} \lambda_f W_f}{\sum_{f=1}^{F_s} \lambda_f}$$
- The average packet delay W_f of each flow $f = 1 \ldots F_s$ depends on the delay suffered by the flow on each link of its routing path(s).
- Next, we consider first no packet loss in any of the network links.

Average packet delay of each flow

For a flow with a single routing path (for example, Flow 1), the average packet delay W_1 is the sum of the average delays suffered by the packets on the links of its routing path.

$$W_1 = W_{A1} + d_A + W_{B1} + d_B$$

W_{A1} – average delay (average queuing delay + average transmission time) of flow 1 in link A

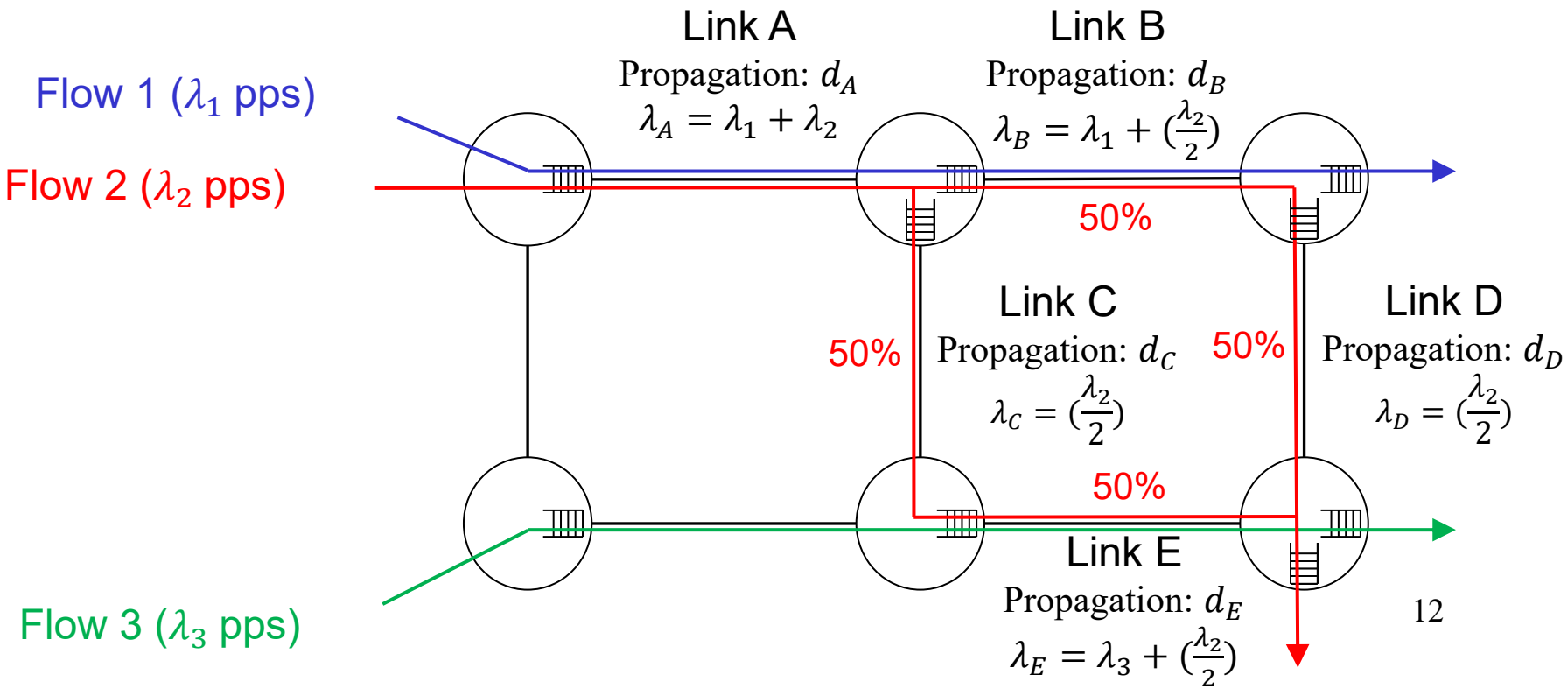


Average packet delay of each flow

For a flow with multiple routing paths (for example, Flow 2), the average packet delay W_2 of the flow is:

- the weighted average of the average packet delay on each routing path,
- the weights are the packet percentages routed through each routing path.

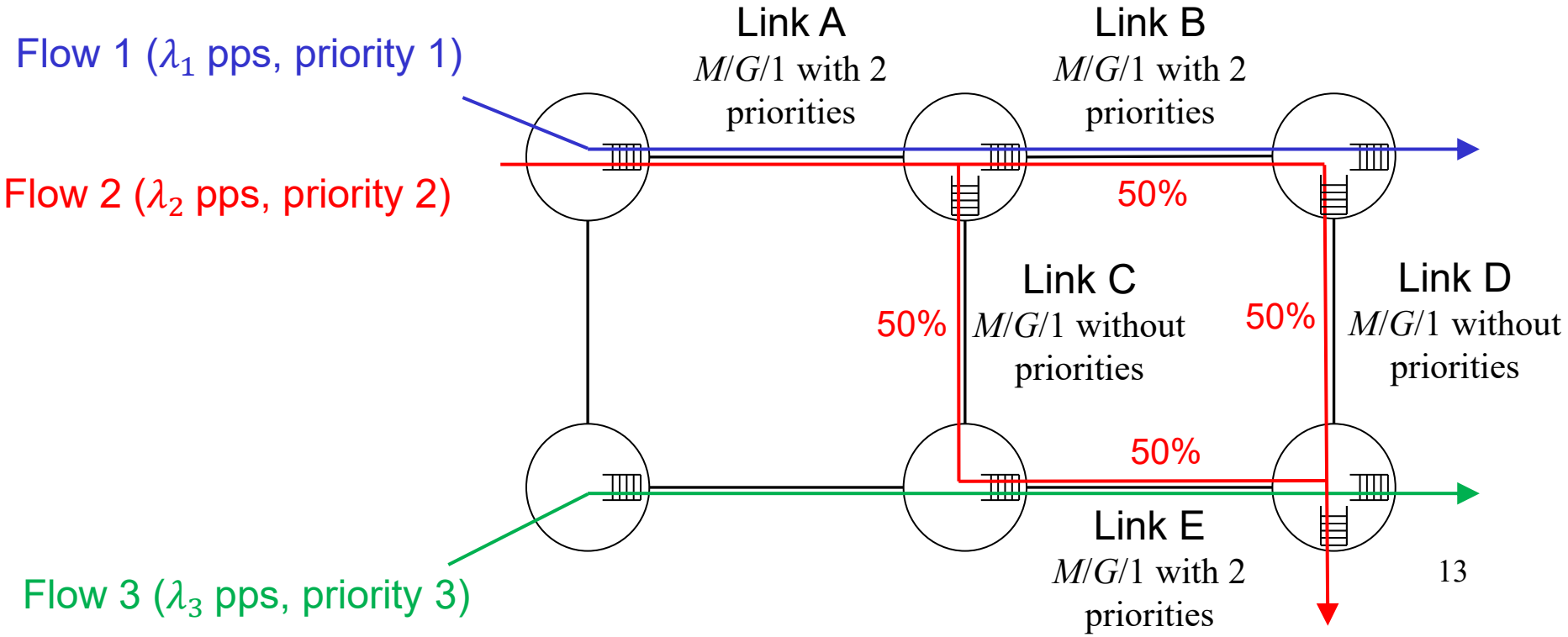
$$W_2 = 0.5\times(W_{A2} + d_A + W_{B2} + d_B + W_{D2} + d_D) + \\ + 0.5\times(W_{A2} + d_A + W_{C2} + d_C + W_{E2} + d_D)$$



Average packet delay of each flow with priority queueing

Consider that each flow is assigned a global (i.e., network-wide) priority:

- Each link is modelled by a *M*/*G*/1 with the priorities considering only the priorities of the flows supported by the link
- In the example below, Flow 2 is the second priority flow in links A and B (Flow 2 has lower priority than Flow 1) but it is the first priority flow in link E (Flow 2 has higher priority than Flow 3)



Packet loss rate of unicast services

- Consider the traffic of a unicast service s defined by a set of flows $f = 1 \ldots F_s$. If for each flow $f = 1 \ldots F_s$:
  - the packet arriving rate at its source node is λ_f (in pps)
  - the packet loss rate of each flow is θ_f

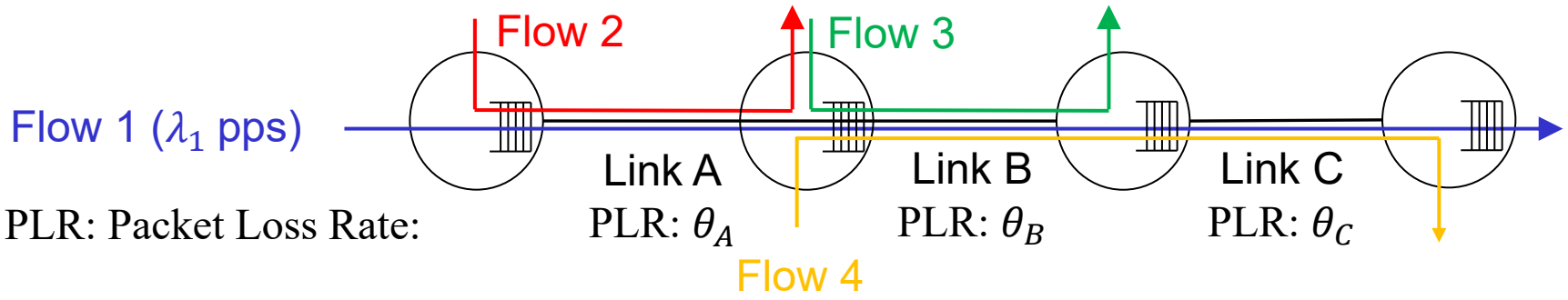
  then, the service packet loss rate θ_s is the weighted average:

$$\theta_s = \frac{\sum_{f=1}^{F_s} \lambda_f \theta_f}{\sum_{f=1}^{F_s} \lambda_f}$$

- Like in the average packet delay case, the packet loss rate θ_f of each flow $f = 1 \ldots F_s$ depends on its routing path(s).
- Recall that packet loss can happen when:
  - the queues in the output interfaces of the links are not enough large
  - the links introduce a significant bit error rate (*ber*)
- When there is packet loss on one link in a routing path, the packet arriving rate in the next links of the routing path decreases.
- So, the average packet delay and packet loss rate suffered by a flow on its routing path(s) are interrelated (next slides).

Average packet delay and packet loss rate

Consider one flow (Flow 1) on a single routing path, whose links are also in the routing paths of other flows (Flow 2, Flow 3 and Flow 4):



- The packet arrival rate of Flow 1 on each link is:

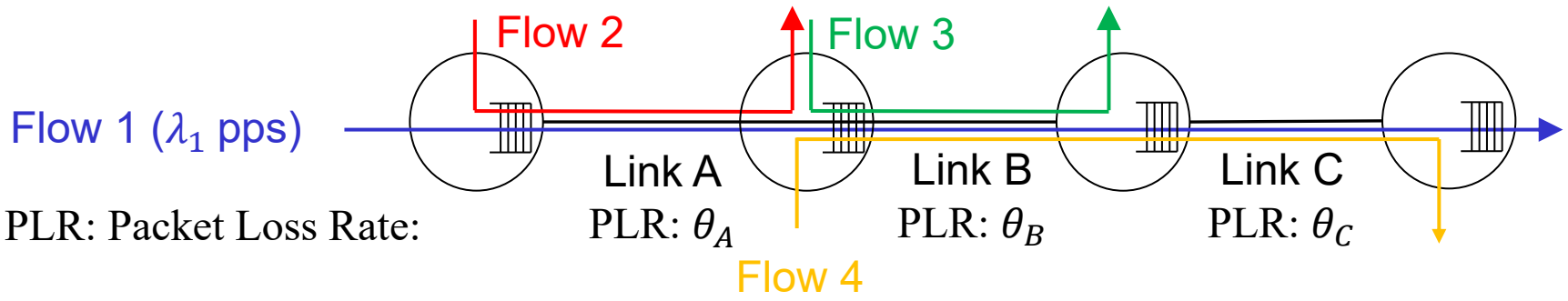
Link A: λ_1 ← the packet rate sent by the origin of the flow

Link B: $\lambda_1 \times (1 - \theta_A)$ ← the packet rate not lost in Link A

Link C: $\lambda_1 \times (1 - \theta_A) \times (1 - \theta_B)$ ← the packet rate not lost neither in Link A nor B

Average packet delay and packet loss rate

Consider one flow (Flow 1) on a single routing path, whose links are also in the routing paths of other flows (Flow 2, Flow 3 and Flow 4):



- The packet loss rate θ_1 suffered by Flow 1 in its routing path is:

$$\theta_1 = \theta_A + (1-\theta_A)\times\theta_B + (1-\theta_A)\times(1-\theta_B)\times\theta_C$$

the packet rate lost in Link A

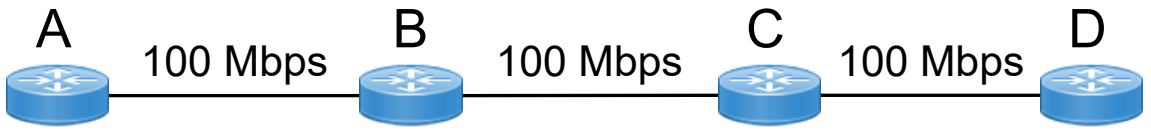
the packet rate not lost in Link A and lost in Link B

the packet rate not lost neither in Link A nor B and lost in Link C

Example 1

Consider the following network supporting one service composed by 3 flows: Flow 1 from A to D with λ_1= 5000 pps, Flow 2 from A to B with λ_2= 2000 pps and Flow 3 from B to D with λ_3= 1000 pps. The packet size of the service is exponentially distributed with average 1250 Bytes.

Link BC has a *ber* of 10^{-5} and Link CD has a *ber* of 10^{-6}. FIFO is the packet scheduler in all links, and all queues are very large.



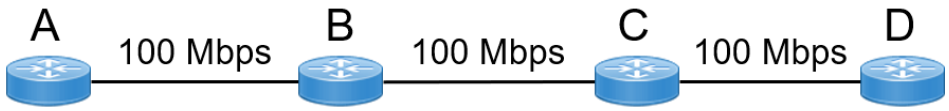
Determine:

(a) the packet loss rate suffered by Flow 1 in the network,

(b) the average packet delay provided by the network to Flow 1.

Example 1 – solution of (a)

Flow 1: A to D, λ_1= 5000 pps
Flow 2: A to B, λ_2= 2000 pps
Flow 3: B to D, λ_3= 1000 pps
Avg. packet size = 1250 Bytes
Link BC: *ber* = 10^{-5} Link CD: *ber* = 10^{-6}



Determine:

(a) the packet loss rate suffered by Flow 1 in the network,

Since all queues are very large, packet loss θ_1 of Flow 1 is only due to *ber* on links BC and CD.

Packet loss of BC: $\theta_{BC} = 1 - (1 - 10^{-5})^{8\times1250} = 0.0952 = 9.52\%$

Packet loss of CD: $\theta_{CD} = 1 - (1 - 10^{-6})^{8\times1250} = 0.00995 = 0.995\%$

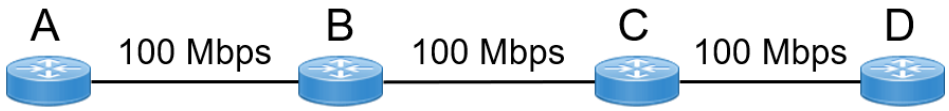
Packet loss rate of Flow 1: $\theta_1 = \theta_{BC} + (1 - \theta_{BC})\times\theta_{CD} = 0.1042 = 10.42\%$

This also means that:

- the packet arriving rate of Flow 1 is $\lambda_1 = 5000$ pps in links AB and BC and $\lambda_1 = 5000\times(1 - 0.0952) = 4524$ pps in link CD
- the packet arriving rate of Flow 3 is $\lambda_3 = 1000$ pps in link BC and $\lambda_3 = 1000\times(1 - 0.0952) = 904.8$ pps in link CD

Example 1 – solution of (b)

Flow 1: A to D, λ_1= 5000 pps
Flow 2: A to B, λ_2= 2000 pps
Flow 3: B to D, λ_3= 1000 pps
Avg. packet size = 1250 Bytes
Link BC: *ber* = 10^{-5} Link CD: *ber* = 10^{-6}



Determine:

(b) the average packet delay provided by the network to Flow 1.

$$\mu_{AB} = \mu_{BC} = \mu_{CD} = \frac{100\times10^6}{8\times1250} = 10^4 \text{ pps}$$

All links are modelled by an *M*/*M*/1

$$W_{ij} = \frac{1}{\mu_{ij} - \lambda_{ij}}$$

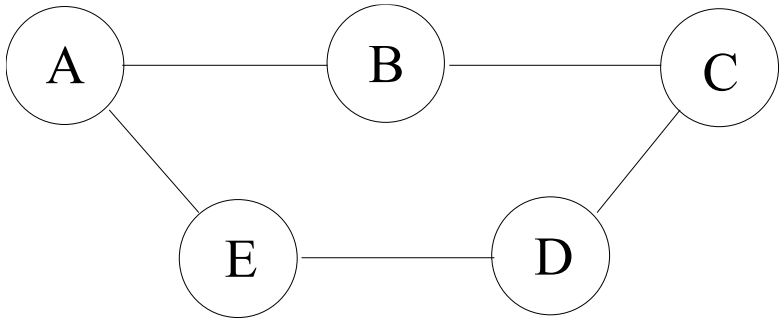
$$W_1 = \frac{1}{\mu_{AB} - \lambda_{AB}} + \frac{1}{\mu_{BC} - \lambda_{BC}} + \frac{1}{\mu_{CD} - \lambda_{CD}}$$

$$= \frac{1}{10^4 - (5000 + 2000)} + \frac{1}{10^4 - (5000 + 1000)} + \frac{1}{10^4 - (5000\times(1 - \theta_{BC}) + 1000\times(1 - \theta_{BC}))}$$

$$W_1 = 8.021\times10^{-4} \text{ sec} = 802.1 \ \mu\text{sec}$$

Calculated in (a)

Example 2



Consider the IP network of the figure where all links are bidirectional with a capacity of 10 Mbps on each direction.

The network is supporting a single service with 4 packet flows:

- from A to C with a Poisson rate of 1000 pps,
- from A to D with a Poisson rate of 250 pps,
- from B to D with a Poisson rate of 1000 pps,
- from B to E with a Poisson rate of 750 pps.

Packet sizes are exponentially distributed with average 500 bytes in all flows. The propagation delay of link B-C is 10 msec. in each direction and neglectable in all other links. The routing protocol is OSPF.

Determine:

(a) the average packet delay of each flow when all OSPF costs are 1,

(b) the average packet delay of the service when all OSPF costs are 1,

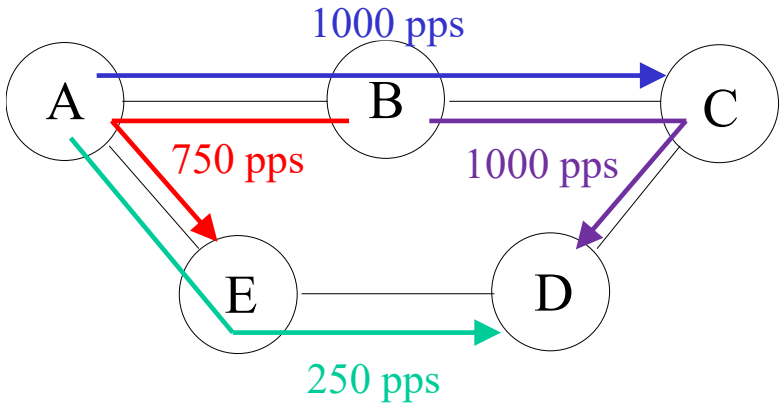
(c) the load (in %) on each direction of each link when all OSPF costs are 1,

(d) the OSPF costs that minimize the highest link load,

(e) the average packet delay of the service in the previous solution.

Example 2 – solution of (a)

- Bidirectional links of 10 Mbps
- Average packet size = 500 bytes
- Propagation delay of link B-C: 10 msec. in each direction



(a) the average packet delay of each flow when all OSPF costs are 1,

$$\mu_{AB} = \mu_{BA} = \mu_{BC} = \cdots = \frac{10\times10^6}{8\times500} = 2500 \text{ pps}$$

All links can be modelled by an *M*/*M*/1

$$W_{ij} = \frac{1}{\mu_{ij} - \lambda_{ij}} + \delta_{ij}$$

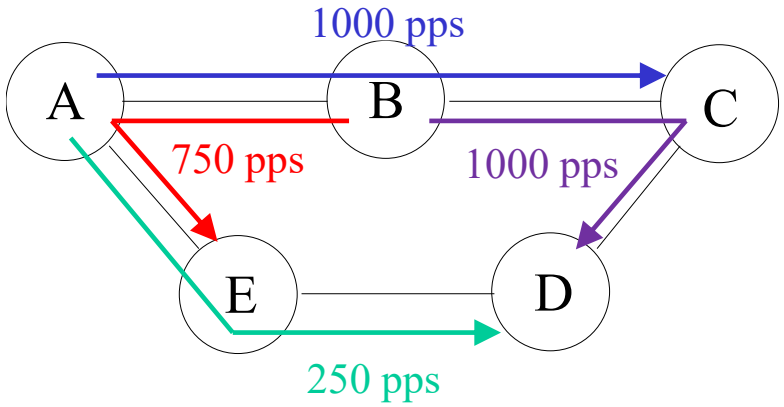
$$W_{A\to C} = \frac{1}{\mu_{AB} - \lambda_{AB}} + \frac{1}{\mu_{BC} - \lambda_{BC}} + \delta_{BC} = \frac{1}{2500 - 1000} + \frac{1}{2500 - (1000 + 1000)} + 0.01 = 0.0127 \text{ sec}$$

$$W_{A\to D} = \frac{1}{\mu_{AE} - \lambda_{AE}} + \frac{1}{\mu_{ED} - \lambda_{ED}} = \frac{1}{2500 - (750 + 250)} + \frac{1}{2500 - 250} = 0.0011 \text{ sec}$$

$$W_{B\to D} = \frac{1}{\mu_{BC} - \lambda_{BC}} + \delta_{BC} + \frac{1}{\mu_{CD} - \lambda_{CD}} = \frac{1}{2500 - (1000 + 1000)} + 0.01 + \frac{1}{2500 - 1000} = 0.0127 \text{ sec}$$

$$W_{B\to E} = \frac{1}{\mu_{BA} - \lambda_{BA}} + \frac{1}{\mu_{AE} - \lambda_{AE}} = \frac{1}{2500 - 750} + \frac{1}{2500 - (750 + 250)} = 0.0012 \text{ sec}$$

Example 2 – solution of (b)



- Bidirectional links of 10 Mbps
- Average packet size = 500 bytes
- Propagation delay of link B-C: 10 msec. in each direction

(b) the average packet delay of the service when all OSPF costs are 1,

The average service packet delay is the weighted average of the avg. packet delay of each flow with weights given by the flows arriving rates:

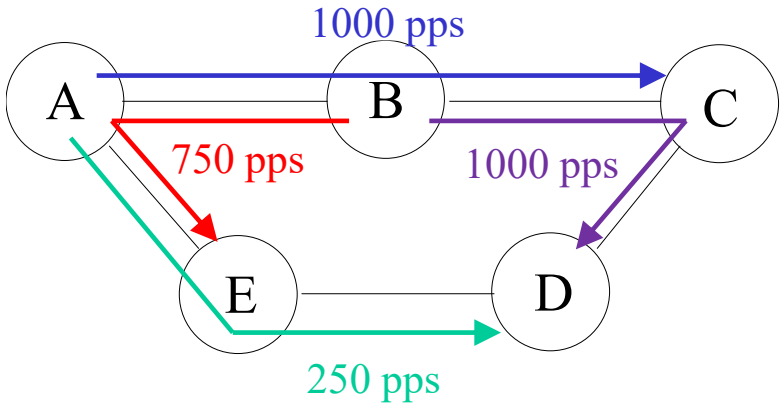
$$W = \frac{\lambda_{A\to C}\times W_{A\to C} + \lambda_{A\to D}\times W_{A\to D} + \lambda_{B\to D}\times W_{B\to D} + \lambda_{B\to E}\times W_{B\to E}}{\lambda_{A\to C} + \lambda_{A\to D} + \lambda_{B\to D} + \lambda_{B\to E}}$$

Calculated in (a)

$$W = \frac{1000\times W_{A\to C} + 250\times W_{A\to D} + 1000\times W_{B\to D} + 750\times W_{B\to E}}{1000 + 250 + 1000 + 750}$$

$$W = 0.00865 \text{ sec} = 8.65 \text{ msec}$$

Example 2 – solution of (c)



- Bidirectional links of 10 Mbps
- Average packet size = 500 bytes
- Propagation delay of link B-C: 10 msec. in each direction

(c) the load (in %) on each direction of each link when all OSPF costs are 1,

$$\mu_{AB} = \mu_{BA} = \mu_{BC} = \cdots = \frac{10\times10^6}{8\times500} = 2500 \text{ pps}$$

$$U_{AB} = \frac{\lambda_{AB}}{\mu_{AB}} = \frac{1000}{2500} = 0.4 = 40\% \qquad U_{BA} = \frac{\lambda_{BA}}{\mu_{BA}} = \frac{750}{2500} = 0.3 = 30\%$$

$$U_{BC} = \frac{\lambda_{BC}}{\mu_{BC}} = \frac{1000+1000}{2500} = 0.8 = 80\% \qquad U_{CD} = \frac{\lambda_{CD}}{\mu_{CD}} = \frac{1000}{2500} = 0.4 = 40\%$$

$$U_{AE} = \frac{\lambda_{AE}}{\mu_{AE}} = \frac{250+750}{2500} = 0.4 = 40\% \qquad U_{ED} = \frac{\lambda_{ED}}{\mu_{ED}} = \frac{250}{2500} = 0.1 = 10\%$$

Example 2 – solution of (d)

- Bidirectional links of 10 Mbps
- Average packet size = 500 bytes
- Propagation delay of link B-C: 10 msec. in each direction

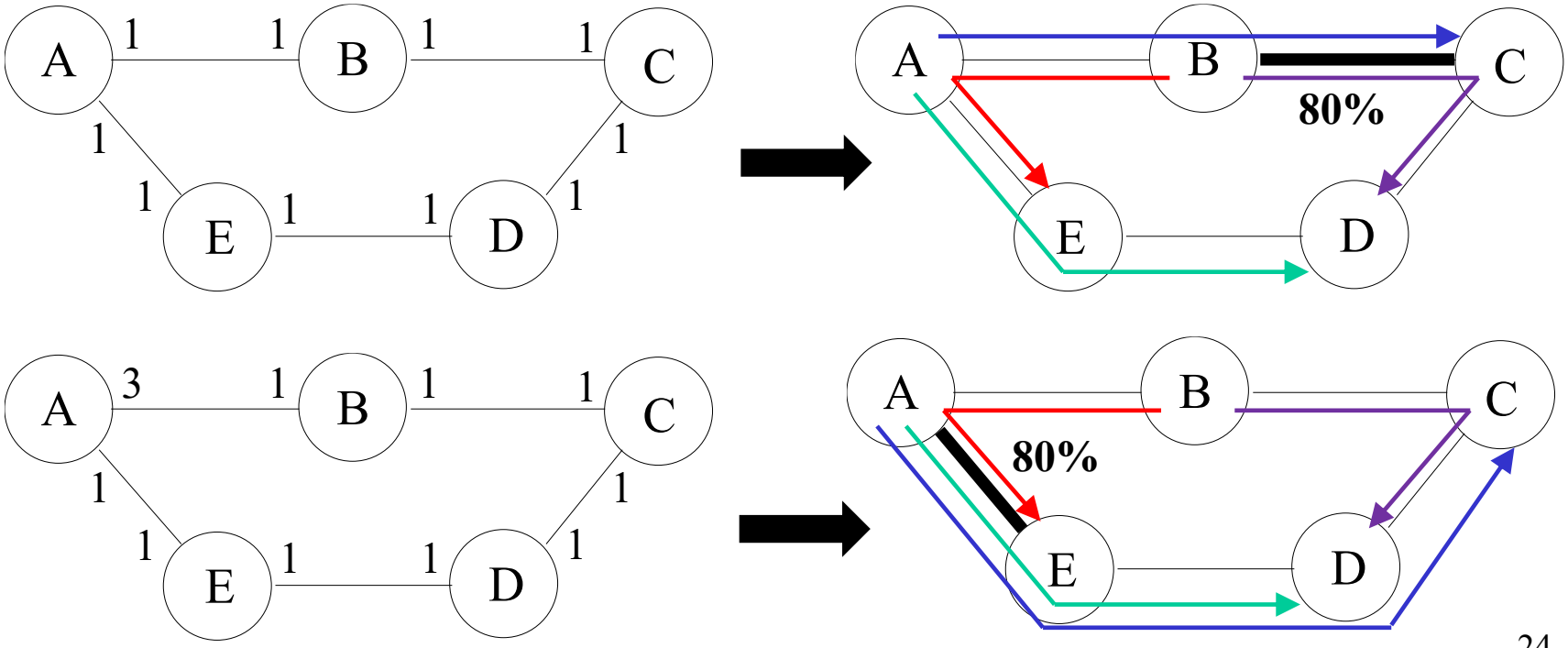
$\lambda_{A\to C} = 1000$ pps

$\lambda_{A\to D} = 250$ pps

$\lambda_{B\to D} = 1000$ pps

$\lambda_{B\to E} = 750$ pps

(d) the OSPF costs that minimize the highest link load,



Example 2 – solution of (d)

- Bidirectional links of 10 Mbps
- Average packet size = 500 bytes
- Propagation delay of link B-C: 10 msec. in each direction

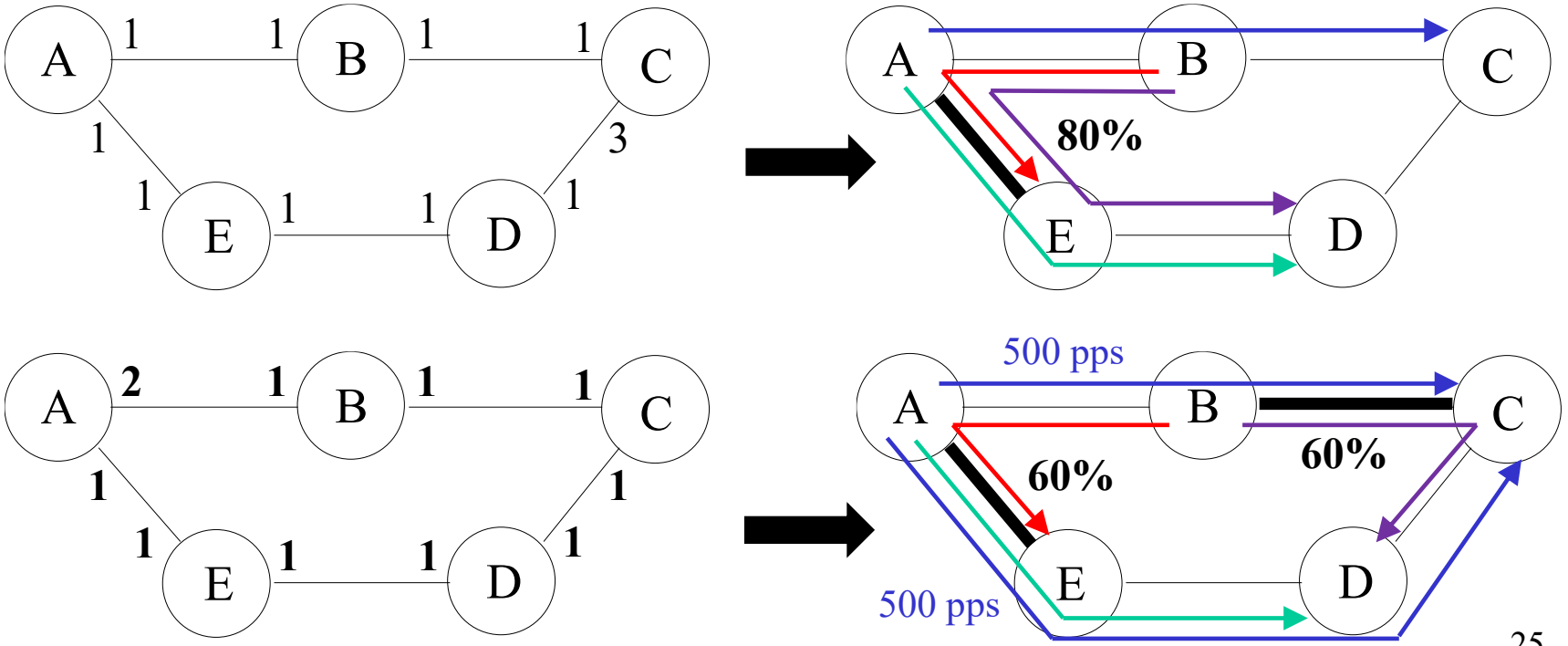
$\lambda_{A\to C} = 1000$ pps

$\lambda_{A\to D} = 250$ pps

$\lambda_{B\to D} = 1000$ pps

$\lambda_{B\to E} = 750$ pps

(d) the OSPF costs that minimize the highest link load,



Example 2 – solution of (e)

- Bidirectional links of 10 Mbps
- Average packet size = 500 bytes
- Propagation delay of link B-C: 10 msec. in each direction

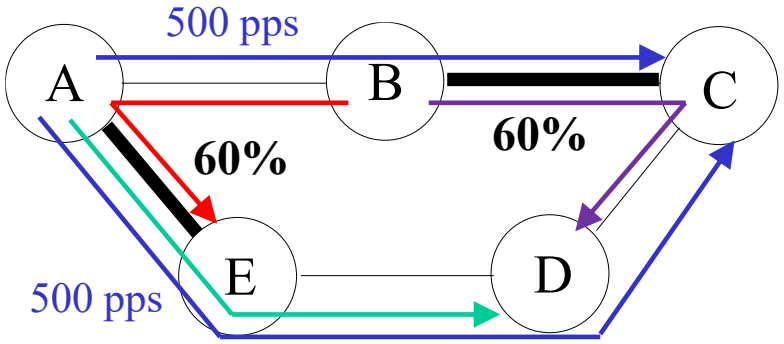
(e) the average packet delay of the service in the previous solution.

$\lambda_{A\to C} = 1000$ pps

$\lambda_{A\to D} = 250$ pps

$\lambda_{B\to D} = 1000$ pps

$\lambda_{B\to E} = 750$ pps



Resolving (e) as it was previously done in (a) and (b):

$$W = 0.00667 \text{ sec} = 6.67 \text{ msec}$$

Recall that in (b): $W = 8.65$ msec

So, the solution computed in (d) has improved both the average service packet delay and the worst link load value.